

Airbag Designer

Grams of a solid to litres of a gas — and what the last step of that chain actually depends on.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/airbag-designer/

1 Before you open anything

Answer from what you already know, on paper, before the screen tells you anything. The point is to have a number to compare against later.

PREDICTION

An inflator holds 100.0 g of sodium azide, which decomposes completely: $2 \text{NaN}_3 \rightarrow 2 \text{Na} + 3 \text{N}_2$. The gas in a deployed bag reaches roughly 300 °C at 1.50 atm. How many litres of N_2 does the bag hold under those conditions? Commit to a number. ($M(\text{NaN}_3) = 65.01 \text{ g/mol}$.)

In one sentence, say what conversion you used for the last step — moles of N_2 to litres — and where that number came from.

2 Reading the chain

Open the demonstration and press **Reset**. The four cards under the bag are one dimensional-analysis chain, left to right. The panels on the right report the same result two ways.

1. Check that **Sodium azide weighed out** reads 100.0 g, **Gas temperature** reads 25 °C, **Pressure in the bag** reads 1.00 atm, and **Driver bag — 60 L** is the selected target.
2. Copy the four card values, in order, into the table.
3. Then copy the four numbers from the two panels on the right.

Where it appears on screen	What it reads
Card 1 – NaN_3 weighed out (g)	
Card 2 – NaN_3 (mol)	
Card 3 – N_2 produced (mol)	
Card 4 – N_2 gas in the bag (L)	
Molar volume (L/mol)	
Gas volume (L)	
Shortcut answer (L)	
Off by (%)	

a. Card 2 to card 3 multiplies by $3/2$. Point at the exact place that fraction comes from – not “the equation” but which two numbers in it.

b. The header on card 4 names a molar volume *and* two conditions. Write the header exactly as it appears, then say why the conditions are printed there at all when cards 1–3 needed no conditions.

c. Switch the target to **Passenger – 150 L** and back, watching your eight numbers. Do any of them change? Say what that tells you about the difference between how much gas you made and how much gas the bag wants.

3 What the last step actually depends on

Keep the mass and the pressure fixed. Only the temperature moves, so the number of moles of N_2 is identical in every row.

1. Press **Reset**. Confirm the mass is 100.0 g and the pressure is 1.00 atm; do not touch either again during this part.

2. Drag **Gas temperature** to each value in the first column.
3. Record **Molar volume** and **Gas volume**. Fill the T (K) and the last column yourself — the demonstration does not print them.

Gas temperature	T (K)	Molar volume (L/mol)	Gas volume (L)	Gas volume ÷ T (K)
0 °C				
100 °C				
200 °C				
300 °C				
400 °C				

a. State the relationship between gas volume and temperature in your own words, in one sentence, and say which temperature scale your sentence is true on.

b. From 100 °C to 200 °C the Celsius reading doubled. By what factor did the gas volume actually grow? Use your T (K) column to account for the difference.

c. Press **Hot deploy** (300 °C, 1.50 atm) and record **Gas volume**: _____ L. Compare it with your part 1 prediction. If they differ, name the specific assumption that caused the gap.

Every molar volume on this page is RT/P for an *ideal* gas. The demonstration says where that stops being safe, in the panel below the chart — read it, including the part about what else is packed into a real inflator alongside the azide.

4 Running the chain backwards

A design question rarely gives you the mass. It gives you the bag and asks what to weigh out. **Weigh out exactly enough** solves the chain in reverse, for the conditions currently set.

1. Press **Reset**, then press **Weigh out exactly enough**. Record the mass and the fill percentage (row 1).
2. **On paper first:** work out how many grams the same 60 L bag needs at 300 °C and 1.50 atm. Show it in the space below. Then press **Hot deploy**, press **Weigh out exactly enough** again, and fill row 2.
3. Select **Passenger — 150 L**, press **Room temperature**, press the button once more, and fill row 3.

Conditions	Target bag	Sodium azide weighed out (g)	Fill of the bag (%)
25 °C, 1.00 atm	60 L		
300 °C, 1.50 atm	60 L		
25 °C, 1.00 atm	150 L		

a. Rows 1 and 2 fill the same bag with the same gas. Why does the hot one need *less* azide? Answer in terms of litres per mole, not in terms of “heat makes it bigger”.

b. Here is the same backwards calculation done with the memorised shortcut: $60 \text{ L} \div 22.4 \text{ L/mol} = 2.679 \text{ mol N}_2$, $\times 2/3 = 1.786 \text{ mol NaN}_3$, $\times 65.01 \text{ g/mol} = 116.1 \text{ g}$. Set **Gas temperature** to 300°C , **Pressure in the bag** to 1.50 atm , **Driver bag** — **60 L** as the target, and drag the mass to **116.1 g**.
Gas volume: _____ L Fill: _____% The verdict says:

The arithmetic in that shortcut calculation contains no mistakes. Say precisely what is wrong with it anyway.

c. Press **Textbook STP** (0°C , 1.00 atm). Record **Off by**: _____%. The panel note changes wording here. Using RT/P , explain why 22.4 L/mol is the right factor at this one setting.

5 Solve these without the demonstration

Close the page or look away. Show your working. Use $R = 0.0821 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$, $M(\text{NaN}_3) = 65.01 \text{ g/mol}$, and the balanced equation $2 \text{NaN}_3 \rightarrow 2 \text{Na} + 3 \text{N}_2$. Then reopen the demonstration, dial in each problem, and check yourself.

1. 45.0 g of NaN_3 decomposes completely. Find the moles of N_2 produced and the volume that gas occupies at 27°C and 1.00 atm .

2. A 150 L passenger bag has to be filled at 150°C and 1.20 atm . What mass of NaN_3 does the inflator need?

3. Two inflators hold identical masses of NaN_3 and both deploy at 1.00 atm — one on a $-20\text{ }^\circ\text{C}$ morning, one at $40\text{ }^\circ\text{C}$. Without calculating either volume, find the factor by which the second bag is larger, and explain why that factor is not $40 \div (-20)$.

4. 100.0 g of NaN_3 is to exactly fill the 60 L driver bag at 1.00 atm. At what temperature, in $^\circ\text{C}$ to the nearest degree, does that happen?

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet used.

- $0\text{ }^\circ\text{C}$ and 1.00 atm is not the only setting where the shortcut is right. Leave the pressure at 1.50 atm and hunt for a temperature that drives *Off by* under 0.5%. Then find another pair. What do all of them have in common?
- The lower chart plots litres per mole across five fixed conditions plus your current setting. Before you move anything, predict where your bar will land relative to the *300 °C, 2.00 atm* row, then set the sliders and see.
- The bag outline is drawn to scale, and width grows as the cube root of volume. Find two settings whose gas volumes differ by a factor of 2 and judge by eye how much wider the second bag is. Check your eye against $2^{1/3}$.
- Select the passenger bag, press **Cold morning**, and press **Weigh out exactly enough**. Watch what the mass slider does at its upper end, and work out what the design would need instead.